

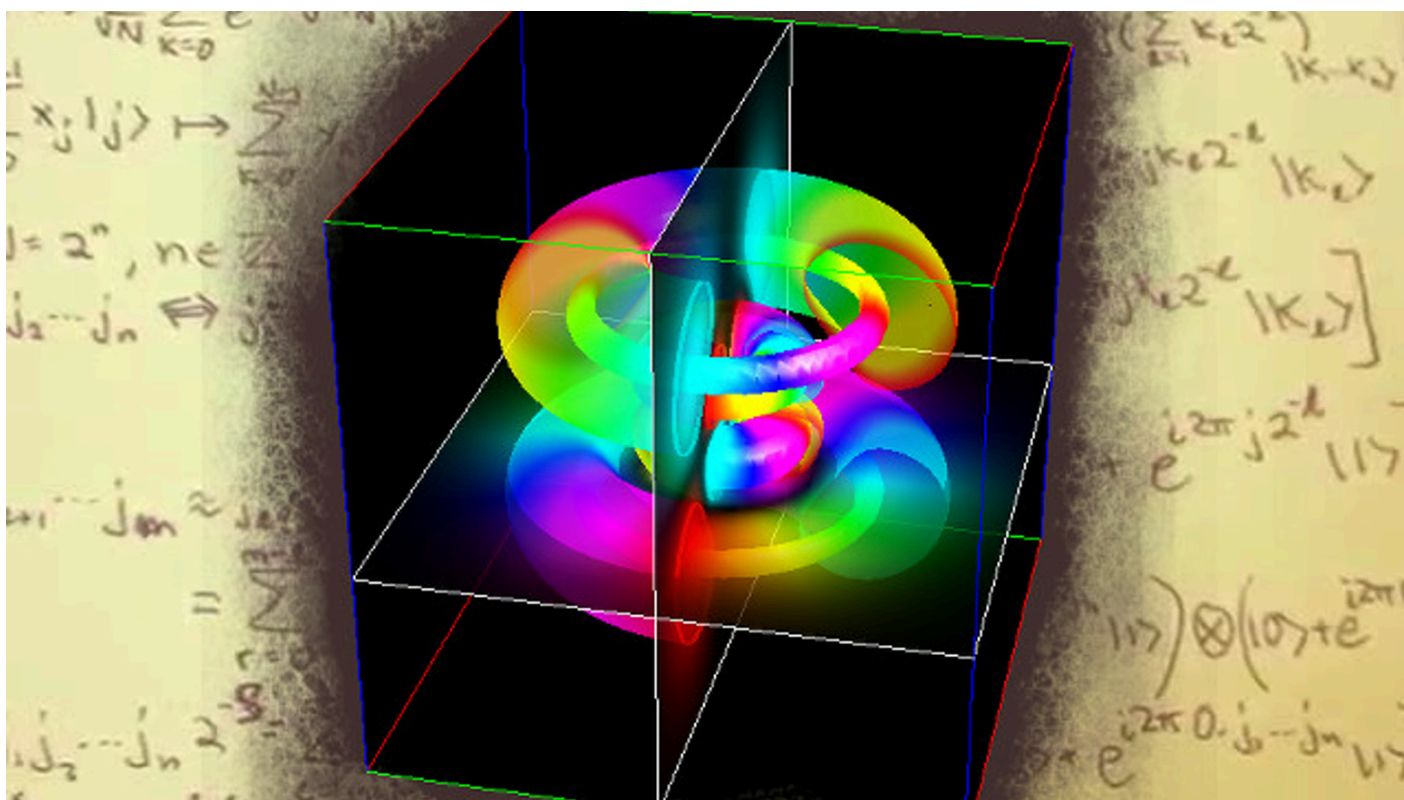
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Quantum WebGL



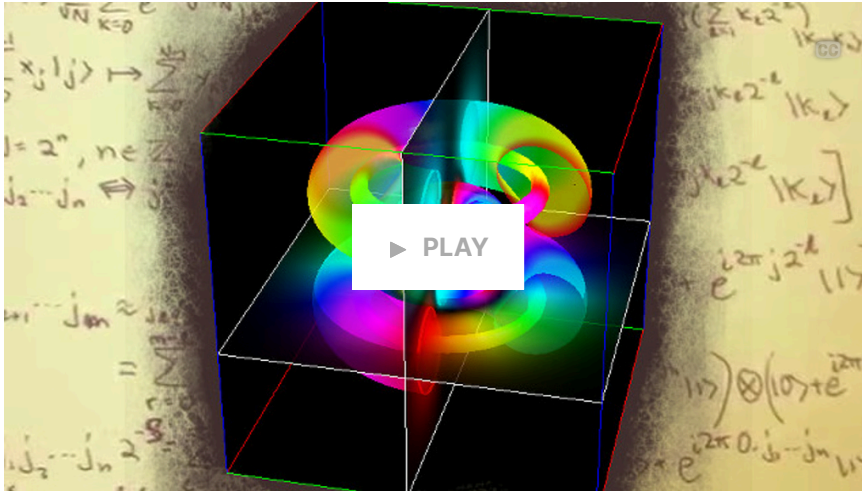
Great academic software is often abandoned as researchers move on. Help save QuantumGL by porting it to the Web (HTML5/WebGL).

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Created by

Henning Dekant

20 backers pledged CA\$ 8,111 to help bring this project to life.

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Funding period

Jun 3 2015 - Aug 2 2015 (60 days)

Quantum WebGL

 [Toronto, Canada](#)  [Software](#)

CA\$ 8,111

pledged of CA\$ 8,000 goal

20

backers

Story

Quantum Mechanics has the reputation of being one of the most difficult subjects to learn. This stems in part from it's counterintuitive predictions but also from the lack of direct experience with the quantum realm. Computer aided visualization can greatly help in this regard and make this science much more accessible.

[Two critically acclaimed books](#) tackled this early in the 21st century. They came with innovative software, included on CD-ROM, to visualize and interactively explore Quantum Mechanics with a great software tool called [QuantumGL](#), which was used, for instance, to [create visualizations of the dynamics of electron orbital states](#).

But as is the case with a lot of great academic software, the researchers moved on, the old QuantumGL program code is no longer maintained, and it became obsolete. While the Windows binary version can still be executed, the Mac OSX version no longer works, nor can it be compiled with the current version of Apple's development tools. Yet, since this software is open source, this hurdle is relatively easy to overcome. But in order to truly bring this software back to life, it should be able to run in a

browser, so that it could be integrated into a MOOCs (Massive Open Online Courses), or any other online learning resource such as [Khan academy](#).

While this will require some substantial reimplementation, the fact that this software uses OpenGL will help immensely, as there is now a web standard, [WebGL](#), that all major browsers support.

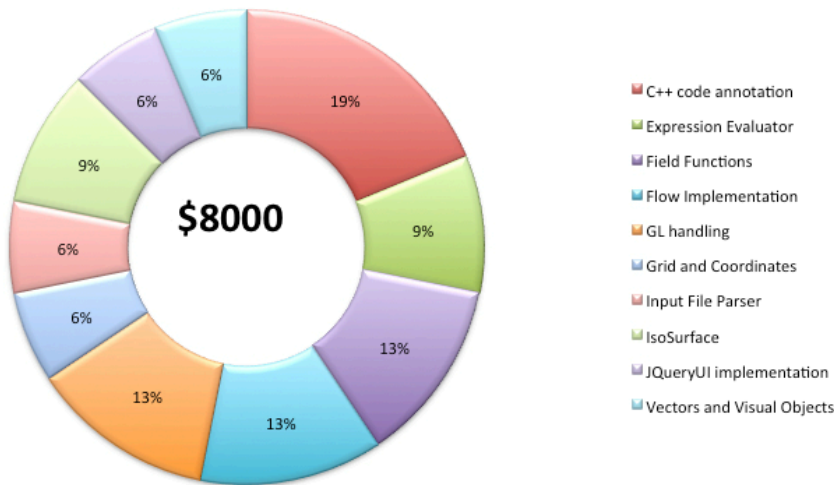
What we want to accomplish is to reimplement the old QuantumGL tool in a way that makes it accessible as an online toolbox, similar to the way that the [Geometry Zen Project](#) allows you to interactively learn about geometric algebra. Our modifications will be made available under the same open license as the original QuantumGL tool, so the full source code including all our enhancements and modernisations will always be available for anyone to use. We will also invite other programmers to add their own modifications, and will offer to help coordinate and distribute their contributions as well.

We will maintain backwards compatibility with the examples contained in the original books, and, instead of being tied to a stand-alone desktop software, remain unshackled, as a web resource that can be used whenever quantum mechanics is taught or illustrated online.

While we want to engage Open Source programmers to contribute to this effort, we don't expect serious coding to go financially unrewarded. The contributions to this project will ensure that we have enough funds to pay freelance Open Source coders. We estimated the breakdown of the necessary activities as depicted in this table and chart:

Why \$8000?

Task	Cost
C++ code annotation	\$1,500
Expression Evaluator	\$750
Field Functions	\$1,000
Flow Implementation	\$1,000
GL handling	\$1,000
Grid and Coordinates	\$500
Input File Parser	\$500
IsoSurface	\$750
jQueryUI implementation	\$500
Vectors and Visual Objects	\$500
Total	\$8,000



Task break-down

Risks and challenges

The amount raised is based on the estimated effort for one developer to complete the project. If we encounter unanticipated technological challenges (e.g. browser incompatibilities) the time line will increase or we will have to make compromises (such as only supporting a specific browser platform).

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